Invariant Offline Constrained Reinforcement Learning for Circular Aquaculture under Heterogeneous Environments

Case Study on AIoT enabled Shrimp Farming in the Mekong Delta

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Abstract. Shrimp farming under circular economy objectives requires policies that maximize profit while satisfying resource constraints across heterogeneous farms and seasons. We study this problem from offline AIoT logs where online exploration is infeasible and distribution shift is unavoidable. We formalize the setting as a Circular Invariant Offline Constrained Markov Decision Process (CIO-CMDP), in which circularity is encoded through a learned monotone utility over episode-level resource metrics. Building on this formulation, we propose IWOC-RL, which integrates an invariant latent world model, a data-driven Circularity Utility Index (CCI), and Lagrangian offline policy optimization. We derive bounds linking out-of-distribution model error and utility approximation error to profit degradation and constraint violations on unseen environments. Experiments on multi-farm shrimp datasets show that IWOC-RL improves OOD profit by 10–12% points, increases circularity scores by 0.06, and reduces violation rates by approximately 2× compared to strong offline RL baselines, demonstrating robust and deployable resource-aware control.

Keywords: Offline Reinforcement Learning · Circular Economy · Aquaculture · Invariant World Model.

1 Introduction

Shrimp farming in the Mekong Delta increasingly relies on AIoT-based monitoring to optimize feeding, aeration, and water exchange under tight economic and environmental constraints [1]. At the same time, circular economy objectives require minimizing water use, energy consumption, and emissions while maintaining profitability [2]. These requirements naturally induce a sequential decision-making problem in which policies must maximize economic return subject to sustainability constraints, across farms and seasons with heterogeneous dynamics.

In practice, control policies must be learned from historical logs, since online exploration is costly and potentially unsafe. This leads to an *offline constrained reinforcement learning* problem under distribution shift [3]. However, existing offline RL and constrained MDP (CMDP) methods typically assume a single stationary environment and optimize step-wise cost signals. They are rarely evaluated under strong cross-environment heterogeneity, and do not explicitly model episode-level sustainability objectives. Meanwhile, circularity indicators in aquaculture are often hand-crafted aggregates of resource metrics [4], rather than learned utilities embedded directly into sequential policy optimization.

These limitations motivate a unified formulation that integrates: (i) multi-environment offline RL under heterogeneous farm–season dynamics, (ii) invariant representation learning for out-of-distribution (OOD) generalization, and (iii) learned episode-level circularity constraints within a CMDP framework.

Research question. How can we learn offline constrained policies that generalize across heterogeneous aquaculture environments while enforcing circularity constraints under distribution shift?

To address this question, we make the following contributions:

- We formalize the *Circular Invariant Offline CMDP* (CIO-CMDP), a multi-environment offline constrained RL framework in which sustainability is encoded through a learned monotone utility over episode-level resource metrics.
- We propose *IWOC-RL*, which integrates an invariant latent world model, a data-driven Circularity Utility Index (CCI), and Lagrangian offline policy optimization for safe control under heterogeneous dynamics.
- We derive OOD generalization bounds that relate world-model error, invariance regularization, and circularity utility approximation error to profit degradation and constraint violations in unseen environments.
- We validate the framework on real multi-farm AIoT shrimp farming data, showing improved OOD profit, higher circularity scores, and reduced violation rates compared with strong offline RL and constrained baselines.

2 Background and Related Work

Offline reinforcement learning and constrained MDPs. Offline RL learns policies from static logs without additional exploration, which is essential in safety-critical domains where online interaction is costly or risky [3]. Representative methods such as Conservative Q-Learning (CQL) [5] and Implicit Q-Learning (IQL) [6] mitigate distribution shift between the behavior policy and the learned policy. Safety requirements are commonly modeled using constrained Markov decision processes (CMDPs), where policies maximize reward subject to auxiliary cost constraints [7]. However, most existing constrained offline RL methods assume a single stationary environment and are rarely evaluated under strong cross-environment heterogeneity.

World models and invariant representations. Model-based RL leverages learned dynamics models for planning or policy optimization in latent space [8]. In parallel, invariant representation learning aims to extract features whose predictive mechanisms remain stable across environments. Invariant Risk Minimization (IRM) formalizes this principle by enforcing a shared optimal predictor across environments [9]. While invariance has improved robustness in supervised learning and some RL settings [10], its integration with offline constrained RL under heterogeneous dynamics remains largely unexplored.

RL in aquaculture and circularity metrics. Reinforcement learning has been applied to aquaculture control in simulated or single-farm settings [11], typically optimizing narrow objectives such as growth or feed efficiency. Circularity and sustainability indicators in aquaculture are usually hand-crafted aggregates of environmental and economic metrics [12] and are not directly embedded as learnable constraints in sequential decision-making. This motivates a unified framework that combines offline RL, invariance, and learned circularity constraints for heterogeneous farm environments.

3 Problem Definition: Circular Invariant Offline CMDP (CIO CMDP)

We study offline constrained RL over a set of heterogeneous aquaculture environments $\mathcal{E}$ (farm$\times$season). Each $e \in \mathcal{E}$ induces an MDP

$$M_e = (\mathcal{S}, \mathcal{A}, P_e, r_e^{\text{prof}}, r_e^{\text{res}}), \tag{1}$$

and the learner observes only an offline dataset collected by logging policies. Circularity is evaluated on episode-level resource metrics m (e.g., water, energy, nutrients) via a monotone utility $C(m)$, and we require $C(m) \geq \tau$. Given training environments $\mathcal{E}_{\text{in}}$ and unseen test environments $\mathcal{E}_{\text{ood}}$, the goal is to learn a stationary policy $\pi(a \mid s)$ that maximizes profit while satisfying the circularity constraint across environments.

CIO CMDP differs from standard CMDPs by (i) enforcing constraints on episode-level aggregated metrics uniformly over multiple environments, and (ii) using a learned monotone utility instead of a fixed observed cost. We assume observations are noisy views of a latent state with stable mechanisms shared across environments; an invariant representation ϕ aims to capture these mechanisms for OOD generalization.

4 Method: IWOC RL (Invariant World model for Offline Constrained RL)

4.1 Invariant World Model

The core of IWOC RL is an **invariant world model**, consisting of an encoder $\phi_\theta : \mathcal{S} \to \mathcal{Z}$ and a dynamics model f_θ in the latent space $\mathcal{Z}$. The encoder maps the observed state s_t to a latent representation $z_t = \phi_\theta(s_t)$, and the dynamics model captures the transition probability $p_\theta(z_{t+1} \mid z_t, a_t, e_t)$, where e_t denotes the environment context (e.g., farm or season).

In our implementation, s_t aggregates water-quality, biomass, and operational signals, while a_t controls aeration/feeding and water exchange. We use a 2-layer MLP encoder ϕ_θ to obtain $z_t \in \mathbb{R}^d$ and a latent Gaussian dynamics model f_θ with auxiliary reward heads, conditioned on an environment embedding e_t concatenated to $[z_t, a_t]$.

The training objective has two parts:

1. A **prediction loss** that minimizes the error between predicted and actual next states (or latent embeddings).
2. An **invariance regularizer** that ensures the learned dynamics in $\mathcal{Z}$ remain consistent across environments. This is achieved by applying an IRM-style gradient penalty or a variance penalty on the conditional dynamics.

Let $\mathcal{L}_{\text{pred}}(\theta)$ be the prediction loss and $\mathcal{L}_{\text{inv}}(\theta)$ the invariance penalty. The **Invariant World Model Objective (IWMO)** is then:

$$\mathcal{L}_{\text{IWMO}}(\theta) = \mathcal{L}_{\text{pred}}(\theta) + \lambda_{\text{inv}}\, \mathcal{L}_{\text{inv}}(\theta), \tag{2}$$

where $\lambda_{\text{inv}} > 0$ controls the invariance regularization. Minimizing $\mathcal{L}_{\text{IWMO}}$ encourages both accurate predictions and cross-environment invariance, facilitating generalization to diverse environments. Section 5 shows that minimizing the IWMO reduces out-of-distribution model error on unseen environments.

4.2 Learning Circularity Utility CCI

We learn a circularity utility $C_\psi(m)$ that maps season-level metrics (e.g., water use, energy, nutrient recovery, profit) to a scalar score, and enforce monotonicity to reflect physically meaningful directions (higher profit and reuse are preferable; higher energy and waste are undesirable). We train C_ψ from expert supervision via pairwise preferences using a margin ranking loss:

$$\mathcal{L}_{\text{CCI}} = \sum_{(i,j)} \max\{0,\ 1 - y_{ij}(C_\psi(m^{(i)}) - C_\psi(m^{(j)}))\}. \tag{3}$$

The learned C_ψ is then used as the constraint signal in offline policy optimization.

4.3 Offline constrained policy optimization

With the invariant world model and learned CCI, we proceed to the **offline constrained policy optimization** step. IWOC RL utilizes a base offline RL algorithm such as Implicit Q Learning (IQL) or Conservative Q Learning (CQL), adapted to handle constraints. The key components of the optimization are:

- A Q function $Q_\omega(z_t, a_t)$ defined in the latent space $\mathcal{Z}$ of the world model.
- A **penalty term** that discourages actions which violate the circularity constraint $C_\psi(m) \geq \tau$, using the predicted circularity scores from the world model and CCI.
- A **conservative regularizer** to prevent policy extrapolation beyond the offline data distribution, reducing the risk of overfitting and unsafe actions.

In a Lagrangian formulation, we can define the constrained objective as

$$\max_{\pi} \; J_{\text{prof}}(\pi) \quad \text{subject to} \quad J_{\text{circ}}(\pi) \triangleq \mathbb{E}\big[C_\psi(m)\big] \geq \tau, \tag{4}$$

where $J_{\text{prof}}(\pi)$ is the profit related return and $J_{\text{circ}}(\pi)$ is the expected circularity score. The corresponding Lagrangian is

$$\mathcal{L}(\pi, \lambda) = J_{\text{prof}}(\pi) + \lambda\big(J_{\text{circ}}(\pi) - \tau\big), \tag{5}$$

with $\lambda \geq 0$ learned together with the policy. In practice, this constraint is implemented as an additional penalty term in the Q learning or actor update.

Additionally, IWOC RL can use the world model for **model based rollouts** to increase sample efficiency. Rollouts are restricted to regions where the world model out of distribution error is estimated to be low, ensuring that the synthetic data generated respects the invariant dynamics learned during training and does not introduce harmful bias.

Overall, IWOC RL couples invariant latent dynamics, a monotone circularity utility, and Lagrangian constrained optimization, enabling the OOD guarantees in Section 5.

Algorithm 1 IWOC RL (high level)

Require: Offline logs $\mathcal{D} = \{(e, s_t, a_t, r_t, m_t, s_{t+1})\}$, target circularity C_{target}, weights $\lambda_{\text{inv}}, \lambda_C$, base offline RL algorithm $\mathcal{A}$ (CQL or IQL)

1: **Train invariant world model:** update θ on $\mathcal{D}$ to minimize $\mathcal{L}_{\text{IWMO}}(\theta) = \mathcal{L}_{\text{pred}}(\theta) + \lambda_{\text{inv}}\mathcal{L}_{\text{inv}}(\theta)$.

2: **Train circularity utility:** construct episode metric vectors m_τ with expert labels or preferences and update ψ by minimizing $\mathcal{L}_{\text{CCI}}$ plus monotonicity and smoothness regularizers.

3: **Train constrained policy:** initialize Q and policy of $\mathcal{A}$; for N_π updates, sample transitions from $\mathcal{D}$ and rollouts from f_θ, compute $r_{\text{prof},t}$ and $C_\psi(m_t)$, define $r'_t = r_{\text{prof},t} - \lambda_C \max(0, C_{\text{target}} - C_\psi(m_t))$, and update Q and the policy using $\mathcal{A}$ on the augmented data.

5 Theoretical Analysis

We now formalize the Circular Invariant Offline CMDP (CIO CMDP) setting and give bounds that connect out of distribution model error and circularity utility approximation error to policy performance.

Setup. Let $\mathcal{E}$ be a finite set of environments corresponding to farm season combinations. For each $e \in \mathcal{E}$ we have a CMDP

$$M_e = (\mathcal{S}, \mathcal{A}, P_e, r_e^{\text{prof}}, r_e^{\text{circ}}, \gamma) \tag{6}$$

with shared state and action spaces and discount $\gamma \in (0, 1)$. For a policy π we denote by $J_e^{\text{prof}}(\pi)$ and $J_e^{\text{circ}}(\pi)$ its discounted profit and circularity returns in environment e. The invariant world model consists of an encoder ϕ_θ, an environment embedding $g(e)$ and a latent dynamics model f_θ that induce a latent kernel

$$q_\theta(z' \mid z, a, e), \tag{7}$$

where $z = \phi_\theta(s)$. The training objective combines a prediction loss and an invariance penalty

$$L_{\text{IWMO}}(\theta) = L_{\text{pred}}(\theta) + \lambda_{\text{inv}} L_{\text{inv}}(\theta), \tag{8}$$

where L_{pred} is the average one step prediction loss over training environments and L_{inv} penalizes variation of $q_\theta(\cdot \mid z, a, e)$ across environments. We write $\mathcal{E}_{\text{in}}$ and $\mathcal{E}_{\text{ood}}$ for the sets of training and OOD environments.

5.1 OOD model error bound

We first relate the model error on OOD environments to the training error and the invariance penalty.

Assumption 1 (Shared latent dynamics up to noise). There exists a reference latent kernel $P_Z^\star$ such that for every $e \in \mathcal{E}$

$$\sup_{z,a} D\big(P_e^Z(\cdot \mid z, a), P_Z^\star(\cdot \mid z, a)\big) \le \delta_{\text{env}} \tag{9}$$

for a divergence D that dominates the training loss. We define the average training and OOD model errors as

$$\varepsilon_{\text{model}}^{\text{train}}(\theta) := \mathbb{E}_{e \in \mathcal{E}_{\text{in}}} \mathbb{E}_{(s,a,s') \sim \mathcal{D}_e} \Big[\ell\big(q_\theta(\cdot \mid \phi_\theta(s), a, e), P_e^Z(\cdot \mid \phi_\theta(s), a)\big)\Big], \tag{10}$$

$$\varepsilon_{\text{model}}^{\text{ood}}(\theta) := \mathbb{E}_{e \in \mathcal{E}_{\text{ood}}} \mathbb{E}_{(s,a,s') \sim \mathcal{D}_e} \Big[\ell\big(q_\theta(\cdot \mid \phi_\theta(s), a, e), P_e^Z(\cdot \mid \phi_\theta(s), a)\big)\Big], \tag{11}$$

where ℓ is the prediction loss.

Assumption 2 (Invariance penalty controls cross environment discrepancy). There exists a constant $c_{\text{inv}} > 0$ such that

$$\mathbb{E}_{(z,a)} D\big(q_\theta(\cdot \mid z, a, e_{\text{ood}}), q_\theta(\cdot \mid z, a, e_{\text{in}})\big) \leq c_{\text{inv}} L_{\text{inv}}(\theta) \tag{12}$$

for any training environment e_{in} and OOD environment e_{ood}.

Theorem 1 (OOD model error bound). *Under Assumptions 1 and 2 there exist constants $c_1, c_2 > 0$ such that for any θ*

$$\varepsilon^{ood}_{model}(\theta) \leq \varepsilon^{train}_{model}(\theta) + c_1 L_{inv}(\theta) + c_2 \delta_{env}. \tag{13}$$

Proof sketch. Add and subtract the in-domain kernel and apply the triangle inequality: $\varepsilon^{\text{ood}}_{\text{model}} \leq \varepsilon^{\text{train}}_{\text{model}}$ + (cross-env kernel gap) + (cross-env predictor gap). The kernel gap is controlled by Assumption 1 and contributes a term proportional to δ_{env}. The predictor gap is controlled by Assumption 2 and contributes a term proportional to $L_{\text{inv}}(\theta)$. Collecting terms yields $\varepsilon^{\text{ood}}_{\text{model}}(\theta) \leq \varepsilon^{\text{train}}_{\text{model}}(\theta) + c_1 L_{\text{inv}}(\theta) + c_2 \delta_{\text{env}}$.

5.2 Policy performance and constraint bounds

We now connect the model error and circularity utility approximation error to profit and constraint satisfaction on OOD environments.

Let $C^\star(m)$ be the true circularity utility of season level metrics m, and let $C_\psi(m)$ be the learned CCI network. Define the uniform approximation error

$$\varepsilon_{\text{CCI}} := \sup_m \big| C_\psi(m) - C^\star(m) \big|. \tag{14}$$

Assumption 3 (Bounded rewards and Lipschitz circularity). Profit and circularity rewards are bounded by $R_{\max}$ and $C_{\max}$. The map $C^\star$ is L_{CCI} Lipschitz in m.

Assumption 4 (Simulation lemma for the world model). There exists a constant H_γ such that for any environment e, policy π and model parameters θ,

$$\big| J^{\text{prof}}_e(\pi) - \hat{J}^{\text{prof}}_e(\pi) \big| \leq H_\gamma \varepsilon^e_{\text{model}}(\theta), \tag{15}$$

and similarly for circularity returns, where $\hat{J}$ denotes the return evaluated inside the learned world model and $\varepsilon^e_{\text{model}}$ is the one step model error in environment e. We consider policies that are approximately feasible on training environments,

$$J^{\text{circ}}_e(\pi) \geq \tau - \xi_{\text{circ}} \quad \forall e \in \mathcal{E}_{\text{in}}. \tag{16}$$

Theorem 2 (OOD profit loss and circularity violation). *Under Assumptions 1 to 4 and the bound in Theorem 1, for any policy π trained using the invariant world model and the learned CCI there exist constants $c_3, c_4 > 0$ such that for every OOD environment $e \in \mathcal{E}_{ood}$*

$$J^{prof}_e(\pi) \geq J^{prof}_{\mathcal{E}_{in}}(\pi) - c_3\big(\varepsilon^{train}_{model}(\theta) + L_{inv}(\theta) + \delta_{env}\big), \tag{17}$$

$$\tau - J^{circ}_e(\pi) \leq \xi_{circ} + c_4\big(\varepsilon^{train}_{model}(\theta) + L_{inv}(\theta) + \delta_{env}\big) + \varepsilon_{CCI}, \tag{18}$$

where $J^{prof}_{\mathcal{E}_{in}}(\pi)$ is the average profit of π over training environments.

Proof sketch. By Assumption 4 (simulation lemma), the return mismatch between the real environment and the learned model is bounded by a constant times the one-step model error. Combining this with Theorem 1 yields the stated OOD profit degradation term proportional to $\varepsilon_{\text{model}}^{\text{train}} + L_{\text{inv}} + \delta_{\text{env}}$. For circularity, apply the same argument to J^{circ}, and then add the CCI approximation error ε_{CCI} to account for using C_ψ instead of $C^\star$.

6 Experiments setup

Data and Setup. The dataset used in this study comes from multiple shrimp farms in the Mekong Delta, with data collected from various sensors and action logs. In total, the dataset contains **18** physical ponds across all farms and **11** production seasons, which correspond to $\mathbf{N}_{\text{epi}} = \mathbf{44}$ season-level episodes (one episode per pond–season) and $\mathbf{T}_{\text{step}} \approx \mathbf{1.2} \times \mathbf{10^5}$ state–action transitions after preprocessing and alignment. We treat each production season as one training episode. We use $\mathbf{N}_{\text{epi-tr}} = \mathbf{26}$ episodes from farms A and B for training, hold out $\mathbf{N}_{\text{epi-val}} = \mathbf{6}$ additional episodes from the same farms for validation and early stopping, and reserve $\mathbf{N}_{\text{epi-ood}} = \mathbf{12}$ episodes from farms C and D as out-of-domain (OOD) test environments. The exact counts per farm and per season are summarized in Table 1.

Evaluation on real logged transitions (no world-model rollouts). All reported in-domain and OOD numbers are computed by *off-policy evaluation* on the held-out logged transitions from the corresponding farms (A/B for in-domain, C/D for OOD). The learned world model is used *only* as a representation/auxiliary training component to learn the policy; we do not simulate OOD trajectories with the world model for reporting, avoiding circular evaluation.

Table 1: Summary of the dataset used in this study.

Farm Type	**No. Farms**	**No. Seasons**	**Sensor Types**	**Data Frequency**
Farm A	5	3	Temperature, pH, DO, Salinity	Hourly
Farm B	4	2	pH, Temperature, Oxygen, FCR	Daily
Farm C	6	4	Temperature, pH, DO, Salinity, Turbidity	Hourly
Farm D	3	2	pH, DO, Temperature, Ammonia	Daily

CCI supervision. Season-level circularity metrics (water use, energy, nutrient recovery, profit) are computed per episode and used to train the CCI via expert thresholds and pairwise preferences (Section 4).

Baselines. We compare IWOC RL against: (i) rule based controllers currently used by farms, (ii) standard offline RL (CQL, IQL), (iii) a constrained offline RL baseline (CMDP) and (iv) IWOC RL variants without invariance or with hand crafted CCI, plus a model free constrained offline RL baseline that applies penalties directly in the action space without a world model.

Evaluation Protocols. We evaluate IWOC RL and the baselines using two performance metrics: **In-domain performance**, where models are trained and tested on the same farms (A and B) to assess performance under known conditions, and **OOD performance**, where models are trained on farms A and B and tested on unseen farms (C or D) to evaluate generalization. The evaluation is based on several metrics including profit, survival rate, feed conversion ratio (FCR), water use per kg of shrimp, energy consumption, emission proxy, Circularity Utility Index (CCI), and constraint violation rate.

Off-policy evaluation (OPE). Since online A/B testing on farms is infeasible, we estimate the performance of a learned policy π from logged data using doubly robust (DR) OPE, which combines importance sampling with a learned Q-function for variance reduction. Let $\mathcal{D}_e = \{(s_t, a_t, r_t, s_{t+1})\}_{t=0}^{T-1}$ be a held-out dataset from environment e, collected by an (unknown) behavior policy $\mu_e(a \mid s)$. We fit a behavior model $\hat{\mu}_e(a \mid s)$ on $\mathcal{D}_e$ and compute clipped importance ratios $\rho_t = \mathrm{clip}\Big(\frac{\pi(a_t|s_t)}{\hat{\mu}_e(a_t|s_t)},\, 0,\, \bar{\rho}\Big)$. The per-trajectory DR estimator is

$$\hat{J}_e^{\mathrm{DR}}(\pi) = \sum_{t=0}^{T-1} \gamma^t \Big(\rho_t\big(r_t + \gamma \hat{V}(s_{t+1}) - \hat{Q}(s_t, a_t)\big) + \hat{V}(s_t)\Big), \tag{19}$$

where $\hat{Q}, \hat{V}$ are learned from the same held-out split using FQE-style regression and early stopping. **Support overlap assumption:** we assume $\pi(a \mid s) > 0 \Rightarrow \mu_e(a \mid s) > 0$ on the support of $\mathcal{D}_e$; in practice we mitigate violations by action-space constraints (policy regularization toward data) and by clipping ρ_t. **Bias–variance:** clipping and behavior-model error can introduce bias, while pure importance sampling has high variance; DR trades these off and is consistent if either $\hat{\mu}_e$ or $(\hat{Q}, \hat{V})$ is well-specified. We report OPE uncertainty via mean $\pm$ std/CI over seeds.

Statistical analysis. We report mean $\pm$ standard deviation over 10 random seeds and additionally provide 95% confidence intervals (CI) computed by a t-distribution across seeds. We use a paired t-test across seeds; improvements are significant at $p < 0.05$ unless stated otherwise. Hyperparameters $(\lambda_{\mathrm{inv}}, \lambda_C)$ are selected on the validation split.

Results. Table 2 reports in domain and OOD performance. IWOC RL achieves the highest profit and CCI with the lowest violation rates in both regimes, and exhibits a smaller OOD performance drop than CQL and IQL. For example, in OOD settings IWOC RL attains 82% profit, 0.90 CCI and 7% violation rate, compared with 70% profit, 0.80 CCI and 15% violation rate for CQL. Figure 1 visualizes the reduced OOD gap and constraint violations.

Table 2: In-domain vs. OOD performance comparison

	In-domain			OOD		
Model	**Profit**	**CCI**	**Viol.**	**Profit**	**CCI**	**Viol.**
IWOC RL	85±2	0.95±0.01	5±1	82±3	0.90±0.02	7±2
CQL	78±3	0.85±0.02	8±2	70±4	0.80±0.03	15±3
IQL	80±2	0.88±0.02	6±1	74±3	0.84±0.02	12±3
Rule-based Ctrl.	65±4	0.75±0.03	12±3	55±5	0.70±0.04	18±4

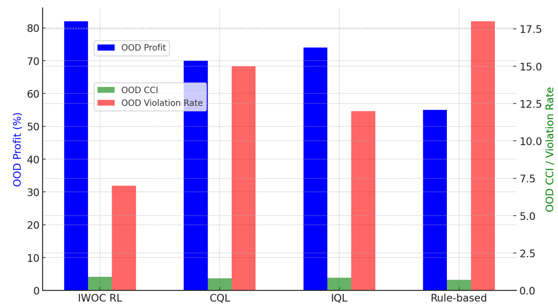

Fig. 1: The OOD performance gap for different models.

Ablation and stress-test studies . Table 3 shows that the invariance regularizer and the learned CCI are both critical for OOD generalization: removing either increases constraint violations and widens the OOD gap, and the model-free constrained baseline performs worst. We additionally construct a stress-test split with larger covariate and reward shift by training on stable seasons from farms A,B and testing on atypical seasons from farms C,D. Under this stronger shift, IWOC RL maintains higher circularity with fewer violations and a smaller profit degradation than baselines.

Table 3: Ablation study (in-domain vs. OOD)

	In-domain			OOD		
Variant	**Profit**	**CCI**	**Viol.**	**Profit**	**CCI**	**Viol.**
IWOC (inv.)	85%	0.95	5%	82%	0.90	7%
IWOC (no inv.)	79%	0.87	9%	75%	0.80	12%
IWOC (HC CCI)	82%	0.91	6%	78%	0.85	10%
MF Constr. RL	70%	0.80	12%	68%	0.75	18%

Isolating invariance effects. To better isolate the contribution of invariance (beyond generic regularization), we compare IWOC RL with additional robustness baselines and report explicit invariance metrics. Specifically, we add: (i) *group DRO* (optimize worst-group training loss over farm/season groups), and (ii) *do-*

main randomization (inject dynamics/observation perturbations during world-model training) as alternative distribution-shift strategies. We further report an *invariance score* computed from the learned representation:

$$S_{\text{inv}} := \frac{1}{|\mathcal{E}_{\text{in}}|(|\mathcal{E}_{\text{in}}|-1)} \sum_{e \neq e'} \mathbb{E}_{(s,a)\sim\mathcal{D}} \left\| \nabla_w \mathcal{L}_e(w \circ \phi_\theta) - \nabla_w \mathcal{L}_{e'}(w \circ \phi_\theta) \right\|_2, \quad (20)$$

where w is the final predictor head and $\mathcal{L}_e$ is the environment-specific one-step prediction loss. Lower S_{inv} indicates better invariance. Finally, we report the OOD *generalization gap* $\varDelta_{\text{OOD}} = J^{\text{prof}}_{\mathcal{E}_{\text{in}}}(\pi) - J^{\text{prof}}_{\mathcal{E}_{\text{ood}}}(\pi)$ for each method to show how invariance reduces transfer degradation.

While we provide OOD bounds and extensive ablations, our empirical study is still limited by (i) the relatively small number of farm–season episodes, (ii) reliance on logged data and off-policy evaluation assumptions, and (iii) potential expert bias in CCI preferences. Future work will expand the dataset, include additional aquaculture regions, and further validate the learned circularity utility with broader expert panels.

Implementation details. We treat each farm-season pair as an environment; farms A,B are used for training/validation and farms C,D for OOD testing. Season-level circularity metrics (water use, energy, nutrient recovery, profit) are computed at the end of each season, normalized to $[0, 1]$, and fed to the monotone CCI network. The encoder and dynamics are 2-layer MLPs (latent dim 64) with an environment embedding; models are trained with Adam and hyperparameters $(\lambda_{\text{inv}}, \lambda_C)$ are selected on validation under the circularity constraint.

CCI learning protocol and label quality. The CCI network C_ψ is trained using expert thresholds for acceptable season-level resource ranges and pairwise preference labels on which season is more circular. We collect **320** comparisons from **5** domain experts across all farms and seasons. Each pair is labeled by at least two experts, with disagreements resolved by majority vote. Inter-rater agreement is substantial (Fleiss' $\kappa = 0.62$). To test robustness, we inject controlled noise into the pairwise labels and find that IWOC RL degrades gracefully up to 20% noisy labels.

7 Discussion and Conclusion

We introduced CIO CMDP and IWOC RL, combining an invariant latent world model with a learned monotone circularity utility for offline constrained control in heterogeneous aquaculture environments. On real AIoT shrimp-farming logs, IWOC RL improves OOD profit and circularity while reducing constraint violations, indicating better transfer under distribution shift. Conceptually, IWOC RL instantiates CIO CMDP by enforcing season-level circularity constraints through a learned utility and invariant dynamics, yielding OOD safety guarantees that link model error and utility approximation to performance and violations. Limitations include dependence on sufficiently diverse environments, potential bias and unobserved confounders in offline logs, and the risk of online deployment. Future work will study safe adaptation and online fine-tuning, and extend the framework to broader circular agriculture and multi-agent settings.

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